

ChatGPT-Based Mental Health Literacy Intervention

A compact pre/post statistical analysis for portfolio demonstration

Executive Summary

Objective. This report evaluates whether a ChatGPT-based mental health literacy intervention was associated with improvements in mental health literacy and related psychological outcomes.

Main result. Total mental health literacy increased significantly from pre-test to post-test (mean difference = 3.84, 95% CI [0.71, 6.97], $p = .017$, Cohen's $d_z = 0.35$).

Supporting pattern. Three of four mental health literacy subscales improved significantly, and mental health self-efficacy showed the strongest exploratory improvement (mean difference = 6.84, $p = .002$, $d_z = 0.46$). Broader emotional outcomes such as affect and flourishing did not show meaningful change.

Interpretation. The findings suggest that the intervention may improve mental health literacy and confidence in handling mental health-related situations, while evidence for broader emotional well-being effects remains limited.

Data Source and Study Context

Dataset. The analysis uses the public Kaggle dataset "AI-Driven Mental Health Literacy", which mirrors the final dataset archived on PsychArchives as "AI-Driven Mental Health Literacy: An Interventional Study from India". The source describes a 20-day ChatGPT-based intervention for college students, with 15-20 minute sessions on alternate days and pre/post measurement of mental health literacy, help-seeking attitude, stigma, self-efficacy, affect, and flourishing. The final sample contains 51 participants.

Portfolio framing. The project is presented as an applied analytics exercise: define an outcome hierarchy, check assumptions, run paired tests, quantify uncertainty and effect size, and communicate findings clearly.

Analytical Method

Design. Pre/post repeated-measures design with the same participants measured before and after the intervention. Because there are only two time points, paired t-tests are the appropriate core inferential method; repeated-measures ANOVA would be equivalent for two measurements and is unnecessary here.

Outcome hierarchy. The primary endpoint is total Mental Health Literacy (MHL_Total). Secondary endpoints are the four MHL subscales. Exploratory endpoints include mental health self-efficacy, help-seeking attitudes, stigma, positive and negative affect, affect balance, and flourishing. Acceptability items are treated descriptively because they are post-test categorical ratings.

Assumptions. Histograms and QQ plots of paired differences were inspected. The primary outcome showed approximate normality and no extreme influential outliers. Secondary and exploratory outcomes showed no severe departures that would invalidate paired t-tests.

Key Statistical Results

Outcome area	Outcome	Mean diff.	95% CI	p	d _z	Interpretation
Primary	MHL Total	+3.84	[0.71, 6.97]	.017	0.35	Significant; small-to-moderate improvement
Secondary	Knowledge of MH Problems	+1.02	[-0.58, 2.62]	.206	0.18	Not significant
Secondary	Erroneous Beliefs / Stereotypes	+0.96	[0.33, 1.59]	.003	0.43	Significant; strongest subscale effect
Secondary	First Aid / Help-Seeking Behaviour	+0.88	[0.03, 1.73]	.042	0.29	Significant; small effect
Secondary	Self-Help Strategies	+0.98	[0.23, 1.73]	.012	0.37	Significant; small-to-moderate effect
Exploratory	Mental Health Self-Efficacy	+6.84	[2.66, 11.03]	.002	0.46	Significant; notable exploratory improvement
Exploratory	Stigma	-2.18	[-4.60, 0.25]	.077	-0.25	Expected direction, not significant
Exploratory	Affect and Flourishing	small	CIs include 0	>.05	near 0	No meaningful change detected

Results Interpretation

Primary outcome. Participants showed a statistically significant increase in total mental health literacy from pre-test ($M = 65.18$) to post-test ($M = 69.02$). The average improvement was 3.84 points, and the 95% confidence interval did not include zero. The effect size ($d_z = 0.35$) indicates a modest but meaningful improvement rather than a large transformation.

Secondary literacy dimensions. The subscale results suggest that the total literacy improvement was mainly driven by changes in erroneous beliefs/stereotypes, first aid and help-seeking behaviour, and self-help strategies. Knowledge of mental health

problems increased by about one point on average but did not reach statistical significance, suggesting more heterogeneous change across participants on this dimension.

Exploratory outcomes. Mental health self-efficacy increased significantly and showed the largest exploratory effect. This is theoretically coherent: improved literacy may be accompanied by greater confidence in managing mental health-related situations. In contrast, help-seeking attitudes, affect, affect balance, and flourishing did not change significantly. Stigma decreased in the expected direction but remained just above the conventional significance threshold.

Acceptability. Post-intervention acceptability ratings were strongly positive. All participants rated the ChatGPT experience as Good or Excellent; 98% rated the overall intervention as Good or Excellent. Most participants reported feeling more knowledgeable about mental health (84%), and most indicated they would use ChatGPT for mental health literacy in the future (73%). Few participants reported concerns that the information was flawed.

Limitations

- The dataset has no control group, so causal claims should remain cautious.
- The sample is small ($n = 51$) and based on college students, limiting generalizability.
- Multiple outcomes were examined; secondary and exploratory findings should be interpreted with caution.
- Acceptability variables are categorical and post-test only, so they were summarized descriptively rather than tested inferentially.
- The exact intervention delivery and prompt structure are not fully reconstructed in this portfolio analysis.

Conclusion

Overall, this analysis provides evidence that the ChatGPT-based intervention was associated with improved mental health literacy and mental health self-efficacy among participants. The strongest story is not broad emotional change, but rather educational and confidence-related improvement. For a portfolio context, the project demonstrates a complete applied workflow: outcome definition, pre/post testing, diagnostic checks, effect-size interpretation, and concise communication of results.

Source Note

Data source: Kaggle dataset "AI-Driven Mental Health Literacy" by mexwell; related PsychArchives record: "Dataset for: AI-Driven Mental Health Literacy: An Interventional Study from India", DOI 10.23668/psycharchives.13284. The analysis in this report is an independent portfolio analysis of the public dataset.